



A Lightweight Hybrid Deep Learning-Based Intrusion Detection System for Detecting Botnet Attacks in IoT Networks

Sirin Mohammed Hejazi ^{a*}, Ahmad Yasser Alshalabi ^b,
Mohammad Hatamleh ^c and Elham Albaroudi ^d

^a Midocean University, UAE.

^b Damascus University, Syria.

^c Edinburgh Napier University, United Kingdom.

^d University of Salford, United Kingdom.

Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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ABSTRACT

The rapid expansion of the Internet of Things (IoT) has introduced an unprecedented number of interconnected devices, creating new opportunities for automation and data exchange but simultaneously increasing the attack surface for cyber threats. Among these, botnet attacks have emerged as one of the most severe threats, capable of compromising massive networks of IoT devices for malicious purposes such as Distributed Denial of Service (DDoS) and data exfiltration. This research proposes a Lightweight Deep Learning-based Intrusion Detection System (DL-IDS)

*Corresponding author: E-mail: 243001213@st.midocean.edu.km, eng.sereenhejazi@gmail.com;

designed to efficiently detect IoT botnet activities with high accuracy and minimal computational overhead.

The proposed model integrates a hybrid Autoencoder–Long Short-Term Memory (LSTM) architecture that captures both spatial and temporal traffic features. The system preprocesses raw network data through feature selection, Min–Max normalization, and class balancing using SMOTE, before training on benchmark datasets such as BoT-IoT and UNSW-NB15. Evaluation results demonstrate that the model achieves an overall accuracy of 99.4%, precision of 99.2%, and AUC score of 0.997, outperforming traditional machine learning and hybrid deep learning baselines (CNN–LSTM, GRU-hybrid, and SVM-Ensemble).

Furthermore, the lightweight design of the model ensures its deployability in real-time edge computing environments. Experimental validation on a Raspberry Pi 4 (8 GB) confirms that the proposed DL-IDS maintains low latency and minimal memory consumption, making it well-suited for IoT gateway and edge-level applications. This study concludes that lightweight deep learning frameworks can effectively balance detection accuracy and computational efficiency, contributing significantly to securing modern IoT infrastructures against evolving botnet threats.

Keywords: *IoT Security; botnet detection; deep learning; Autoencoder–LSTM; Intrusion Detection System (IDS); anomaly detection; edge computing; BoT-IoT Dataset; UNSW-NB15 dataset; lightweight model; cybersecurity.*

1. INTRODUCTION

1.1 Background of the Study

Significant technological advancements, particularly in computer networking, have ushered in the age of wireless communication and pervasive interconnectivity. Among these innovations, the Internet of Things (IoT) stands out as a transformative paradigm that enables physical devices and objects to communicate, collect, and exchange data through the Internet. This technological convergence is reshaping daily life, industrial operations, and critical infrastructures.

According to Statista, the number of connected IoT devices is projected to exceed 32.1 billion by 2030, nearly doubling from 15.9 billion in 2023 (Vailshery, 2024). While IoT devices span multiple sectors — including healthcare, manufacturing, agriculture, and smart cities — the consumer segment accounts for approximately 60% of all IoT devices. The widespread adoption of IoT promises enhanced efficiency, automation, and quality of life.

However, the same interconnectivity that fuels IoT innovation also introduces significant cybersecurity challenges. The IoT ecosystem is characterized by heterogeneous architectures, resource constraints, and a lack of standardized security protocols (Al-Qaseemi et al., 2016). Additionally, Lara et al. (2020) emphasize that IoT networks often suffer from low capacity and inadequate manual controls, leaving them susceptible to exploitation.

One of the most pervasive threats in IoT environments is the botnet attack — a network of compromised devices controlled remotely by an attacker (often referred to as a *BotMaster*). Botnets are frequently leveraged for malicious activities such as Distributed Denial of Service (DDoS) attacks, phishing, and malware distribution (Srinivasan & Deepalakshmi, 2023).

The release of the Mirai botnet source code in 2016 drastically escalated these threats, enabling even unskilled actors to deploy customized variants for large-scale cyberattacks (Zhang et al., 2020).

To counter such threats, Artificial Intelligence (AI)—particularly Machine Learning (ML) and Deep Learning (DL)—has been extensively explored for intrusion detection. Xiao et al. (2018) assert that AI-driven intrusion detection systems (IDS) can identify anomalous network behaviors in real time. However, these systems often face generalization issues across diverse IoT environments and remain vulnerable to adversarial manipulation.

1.2 Problem Statement

IoT devices are especially vulnerable to botnet attacks due to several inherent weaknesses:

- 1. Insecure Communication Protocols:** Many IoT devices operate over diverse communication standards, often without robust encryption. This creates multiple points of vulnerability across the network (Vardakis et al., 2024).

2. **Default and Weak Credentials:** A majority of IoT devices are shipped with default passwords, which are seldom changed, making unauthorized access trivial for attackers.
3. **Insecure Network Environments:** IoT systems often transmit data over unsecured or poorly configured networks, enabling attackers to intercept and manipulate communications (Mukhtar et al., 2023).
4. **Resource Constraints:** Limited computational capacity and memory make it difficult to implement complex security mechanisms directly on IoT devices.

Traditional signature-based IDS are insufficient against sophisticated and evolving botnets. Although AI-based systems promise enhanced detection accuracy, they often introduce computational overheads unsuitable for lightweight IoT environments (Alwahedi et al., 2024).

Furthermore, Goodfellow et al. (2018) highlight that deep neural networks themselves can be exploited through adversarial attacks, compromising IDS integrity. Additionally, IoT traffic patterns are highly variable, complicating the design of adaptive detection systems (Bakhsh et al., 2023).

Therefore, there is a pressing need for a lightweight yet robust intrusion detection system specifically designed for IoT networks, capable of real-time detection while minimizing resource consumption.

In consumer IoT—particularly smart homes—users often lack cybersecurity awareness, leading to weak passwords, unpatched firmware, and insecure remote access configurations (Vardakis et al., 2024). This human factor, combined with device heterogeneity and limited computational resources, creates an ideal environment for botnet propagation.

1.3 Aim of the Study

The primary aim of this study is to design and develop a lightweight Deep Learning-based Intrusion Detection System (DL-IDS) capable of detecting and mitigating botnet attacks in IoT networks efficiently and in real time, while

minimizing computational and memory resource consumption.

1.4 Research Objectives

The study seeks to achieve the following specific objectives:

1. To investigate how botnets exploit vulnerabilities within IoT systems to launch attacks.
2. To design and develop a lightweight Deep Learning-based Intrusion Detection System (DL-IDS) for IoT networks.
3. To evaluate the performance of the proposed system using benchmark IoT botnet datasets.
4. To optimize the IDS for real-time performance in resource-constrained IoT environments.

1.5 Research Questions

Based on the above objectives, the study will address the following research questions:

1. How do botnets exploit existing vulnerabilities in IoT networks?
2. What Deep Learning approaches are most suitable for developing a lightweight IDS for IoT devices?
3. How effective is the proposed DL-IDS in accurately detecting botnet attacks in IoT environments?
4. To what extent can computational and memory efficiency be optimized without compromising detection accuracy?

1.6 Justification / Rationale of the Study

Despite the increasing prevalence of IoT systems across all sectors, their security remains a major challenge. Traditional IDS solutions are often too resource-intensive for IoT environments, while existing Deep Learning approaches lack adaptability and consume excessive power and memory. Addresses the critical need for on-device anomaly detection in smart-home gateways, where cloud-dependent IDS introduce latency and privacy risks (Vardakis et al., 2024).

This study is justified on three main grounds:

- **Technical Justification:** It contributes to the development of a scalable, efficient, and lightweight IDS suitable for low-power IoT devices.
- **Practical Justification:** The outcomes will enhance IoT device manufacturers' ability to integrate intelligent, embedded security solutions.
- **Academic Justification:** It bridges a critical research gap by combining Deep Learning and lightweight architecture principles to achieve both performance and efficiency.

1.7 Significance of the Study

This research holds significance for multiple stakeholders:

- **For Academia:** It expands the body of knowledge in Deep Learning-based cybersecurity for IoT, offering a replicable model for lightweight IDS design.
- **For Industry:** IoT manufacturers, service providers, and cybersecurity firms can apply the proposed model to safeguard IoT ecosystems cost-effectively.
- **For Policy and Standards:** Insights from this study can inform policy discussions around IoT security frameworks, standardization, and resilience.

The proposed model also supports real-time threat mitigation, which is crucial for emerging fields such as smart healthcare, autonomous systems, and critical infrastructure protection.

1.8 Scope and Limitations of the Study

The study focuses exclusively on IoT botnet detection using Deep Learning techniques. It examines datasets such as Bot-IoT and UNSW-NB15, which are representative of IoT network traffic. The study assumes network traffic originates from typical smart-home devices (e.g., cameras, thermostats, plugs) and may not generalize to industrial or medical IoT without retraining.

Limitations include:

- The study does not address non-botnet-related IoT attacks (e.g., insider threats or physical tampering).

- Model evaluation is based on simulated datasets, not live IoT traffic, though they closely resemble real-world conditions.
- The implementation targets small to medium IoT environments; large-scale deployments may require further optimization.

1.9 Organization of the Study

This research is organized into five chapters as follows:

- **Chapter One – Introduction:** Provides the study background, problem statement, objectives, research questions, significance, and scope.
- **Chapter Two – Literature Review:** Reviews previous studies on IoT security, botnet detection methods, and Deep Learning-based IDS approaches.
- **Chapter Three – Research Methodology:** Describes the datasets, experimental design, DL architecture, and evaluation metrics used.
- **Chapter Four – Results and Discussion:** Presents the findings, model performance analysis, and comparative evaluation with existing IDS frameworks.
- **Chapter Five – Conclusion and Recommendations:** Summarizes the study's outcomes, discusses implications, and provides directions for future research.

2. LITERATURE REVIEW

2.1 Introduction

The Internet of Things (IoT) has rapidly evolved into one of the most transformative technological paradigms of the 21st century. With billions of interconnected devices, IoT enables intelligent sensing, automation, and control across domains such as smart homes, healthcare, transportation, and industrial systems. However, this connectivity exposes networks to an expanding attack surface. Among the most significant threats are botnet attacks, which exploit IoT vulnerabilities to coordinate large-scale cyber intrusions.

This chapter critically reviews scholarly literature related to IoT security and botnet detection, with a specific focus on Deep Learning (DL)-based Intrusion Detection Systems (IDS). The review covers the architecture of IoT networks, types of attacks, existing detection approaches, and gaps in current research.

Recent studies have also highlighted the growing intersection between artificial intelligence, data sovereignty, and sustainable digital transformation within national frameworks such as Saudi Arabia's Vision 2030. For example, Albaroudi et al. (2025a, 2025b, 2025c) discussed AI-driven innovations in digital health, water management, and smart energy systems, underscoring the importance of ethical governance and sustainability. These works, alongside broader investigations into generative AI and algorithmic bias (Albaroudi et al., 2024a; 2024b; 2025d; 2025e), provide relevant context for lightweight and energy-efficient AI applications such as the proposed IoT intrusion-detection framework.

2.2 Internet of Things (IoT) Architecture and Vulnerabilities

The IoT architecture comprises interconnected physical devices—ranging from sensors to embedded systems—that collect and exchange data via wired or wireless communication protocols. These devices typically communicate using lightweight protocols such as 6LoWPAN, ZigBee, Bluetooth Low Energy (BLE), Z-Wave, and Near Field Communication (NFC) (Al-Sarawi et al., 2017). The heterogeneity of these protocols introduces complexity in maintaining secure communication channels.

Due to the absence of standardized security measures and the resource-constrained nature of many IoT devices, attackers often exploit software vulnerabilities, weak authentication mechanisms, and insecure firmware updates. Research by Alghazzawi et al. (2021) highlights that over 70% of IoT devices are vulnerable to at least one security flaw. Similarly, Makhdoom et al. (2023) emphasize that once an IoT device is compromised, attackers can propagate malware across the network to form distributed botnets.

These vulnerabilities are exacerbated by the exponential growth of devices. As Vailshery (2024) projects, IoT connections will exceed 32 billion by 2030, amplifying the potential attack vectors exponentially.

Integrate layered threat model from Vardakis et al. (2024):

- **Perception Layer:** Eavesdropping, side-channel attacks → mitigated by encrypted sensing.
- **Network Layer:** DoS, MITM → mitigated by secure routing (e.g., SCOTRES, WARDOG).
- **Application Layer:** Phishing, malware → mitigated by MFA, code analysis.

2.3 Botnet Attacks in IoT Environments

A botnet refers to a network of compromised devices (bots) controlled remotely by a botmaster through command-and-control (C&C) servers. Botnets enable large-scale cyberattacks such as Distributed Denial of Service (DDoS), data exfiltration, and spam distribution (Iftikhar et al., 2020).

The Mirai botnet, launched in 2016, remains a landmark incident demonstrating IoT vulnerabilities. It exploited weak default credentials to compromise millions of devices, subsequently launching one of the largest DDoS attacks against the DNS provider Dyn, affecting platforms like Twitter, Netflix, and GitHub (Affinito et al., 2023). Since then, variants such as Bashlite, Hajime, and Medusa have evolved, demonstrating adaptive attack behaviors and enhanced persistence.

Recent research by Jin et al. (2024) introduced the *IoT Targeting-Threat Modeling (I3TM)* framework, which characterizes the attack stages of IoT botnets. This framework underscores that many attacks exploit the convergence of operational technologies (OT) and information technologies (IT), blurring the boundary between device-level and network-level security.

2.4 Botnet Detection Techniques in IoT Networks

Early detection of botnet activity is critical to preventing large-scale IoT disruptions. Botnet detection techniques can be broadly categorized into signature-based, anomaly-based, and hybrid approaches.

1. **Signature-Based Detection:** These systems rely on predefined attack patterns. Although efficient for known threats, they

fail against zero-day or evolving botnets (Ali et al., 2020).

2. **Anomaly-Based Detection:** Anomaly-based systems identify deviations from normal behavior. They are suitable for detecting unknown threats but may produce high false positives (Shareena et al., 2021).
3. **Hybrid Systems:** Hybrid approaches combine the advantages of both anomaly and signature-based methods to improve accuracy and reduce false alarms (Soe et al., 2020).

Machine Learning (ML) techniques such as Support Vector Machines (SVM), Decision Trees (DT), Random Forests (RF), and Naïve Bayes (NB) have been traditionally employed for anomaly detection in IoT environments. However, these models struggle to adapt to the nonlinear and high-dimensional nature of IoT traffic.

2.5 Deep Learning Approaches for Botnet Detection

Deep Learning (DL) techniques have emerged as powerful alternatives due to their ability to automatically extract complex patterns from large datasets. Common DL architectures used in IoT botnet detection include:

- **Convolutional Neural Networks (CNNs):** Capture spatial correlations in network data, particularly effective for traffic image representation.
- **Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM):** Capture temporal dependencies and sequential patterns in traffic behavior.
- **Autoencoders:** Provide unsupervised feature learning and anomaly detection by reconstructing input patterns.

Shareena et al. (2021) employed a Deep Neural Network (DNN) trained on the BoT-IoT dataset, achieving an accuracy of 94%. However, DNN models are computationally expensive for IoT edge devices. To address this, Catillo et al. (2023) proposed a *cross-device lightweight DL model*, achieving an F1-score of 0.9998 with minimal resource usage.

Hybrid models that combine multiple DL techniques have demonstrated further promise. Mutambik (2024) integrated CNN and LSTM to leverage both spatial and temporal features, enhancing robustness and adaptability. Similarly, Alkahtani & Aldhyani (2021) reported that CNN–LSTM architectures achieved over 98% accuracy while maintaining low false positive rates.

Recent research trends also explore federated learning (FL) for distributed IDS deployment. Zhang et al. (2024) demonstrated that FL-based IoT IDS can detect botnets collaboratively across devices while preserving data privacy.

2.6 Lightweight Intrusion Detection Systems (IDS) for IoT

A major challenge in DL-based IDS design is resource efficiency. IoT devices typically have limited computational power, memory, and energy capacity, making it difficult to deploy traditional DL models. Therefore, researchers have focused on lightweight architectures that optimize model size and inference latency.

Catillo et al. (2023) introduced a cross-device DL model that trains once using generic IoT traffic data, allowing transferability across different devices. Similarly, Bakhsh et al. (2023) proposed a resource-aware IDS using dimensionality reduction to minimize memory usage while maintaining detection accuracy.

Emerging studies also explore TinyML and Edge AI paradigms, where lightweight DL models are deployed directly on embedded hardware (e.g., Raspberry Pi, ESP32). Singh & Joshi (2025) developed an attention-based LSTM model achieving high detection performance with reduced computational cost, highlighting the potential of micro-optimized AI for IoT security.

While lightweight IDS detect anomalies, they must be paired with secure authentication to prevent impersonation. Lara et al. (2020) demonstrate that even minimal cryptographic protocols (e.g., hash-based key pools) can significantly reduce attack surface in resource-constrained M2M communications—suggesting that future IDS should integrate with lightweight auth at the edge.

2.7 Research Gaps

Despite significant progress, several gaps persist in current research:

1. **Computational Overhead:** Many existing DL models remain too resource-intensive for real-time IoT deployment.
2. **Generalization Limitations:** Models trained on one dataset often underperform on another due to variations in network topology and traffic distribution.
3. **Lack of Adaptability:** Few studies have addressed model adaptability to evolving botnet behaviors and dynamic IoT environments.
4. **Limited Real-World Testing:** Most IDS frameworks are validated on public datasets rather than live IoT environments, raising concerns about scalability and robustness.
5. **Lack of Human-Centric Design:** Most DL-IDS ignore user behavior patterns (e.g., usage schedules), which could reduce false positives in smart homes (Vardakis et al., 2024).
6. **Insufficient Integration with Lightweight Auth:** Few studies co-design IDS with authentication protocols, leaving systems vulnerable to spoofed traffic that bypasses detection.

These gaps highlight the need for a lightweight, adaptive, and high-accuracy IDS that can generalize across IoT networks while maintaining computational efficiency.

2.8 Summary

This chapter has reviewed the literature surrounding IoT vulnerabilities, botnet attack mechanisms, and Deep Learning-based intrusion detection approaches. The analysis reveals that while numerous models achieve high accuracy, most are not optimized for deployment in resource-constrained IoT environments.

The proposed study addresses this research gap by developing a lightweight Deep Learning-based IDS that combines anomaly detection with resource optimization, ensuring real-time protection against botnet attacks without compromising efficiency.

3. RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted in the design, implementation, and evaluation of the proposed Lightweight Deep Learning-based Intrusion Detection System (DL-IDS) for detecting botnet attacks in IoT networks. The

section outlines the research design, data sources, preprocessing procedures, model development, evaluation metrics, and ethical considerations. The aim of this methodology is to ensure that the system achieves a balance between detection accuracy and computational efficiency, enabling its deployment in resource-constrained IoT environments.

3.2 Research Design

The study employs an experimental quantitative research design, which is suitable for developing and evaluating a Deep Learning model through measurable parameters such as accuracy, precision, recall, and F1-score. The design involves the following steps:

1. **Data Acquisition** – Obtaining benchmark IoT botnet datasets.
2. **Data Preprocessing** – Cleaning, balancing, and normalizing the dataset for optimal model performance.
3. **Model Design** – Developing a lightweight DL architecture combining *Autoencoder* and *Long Short-Term Memory (LSTM)* layers.
4. **Training and Testing** – Dividing the dataset into training, validation, and testing subsets.
5. **Performance Evaluation** – Assessing the model using established metrics and comparing results with existing IDS models.

This design allows for systematic experimentation, reproducibility, and objective analysis of results.

Threat Model Alignment: The proposed DL-IDS is evaluated against threats common in smart-home IoT (Vardakis et al., 2024), including DDoS, reconnaissance, and exploit-based botnet activities. The model assumes the attacker can eavesdrop, inject, or replay traffic but cannot physically tamper with the gateway running the IDS.

3.3 Research Framework

The proposed framework integrates Deep Learning techniques to identify and classify botnet activity within IoT network traffic.

Fig. 1 has been revised to clearly illustrate the interconnections among the data collection, preprocessing, deep learning, and evaluation

layers. Each arrow now explicitly represents the data flow and transformation between components, ensuring conceptual transparency in the overall framework.

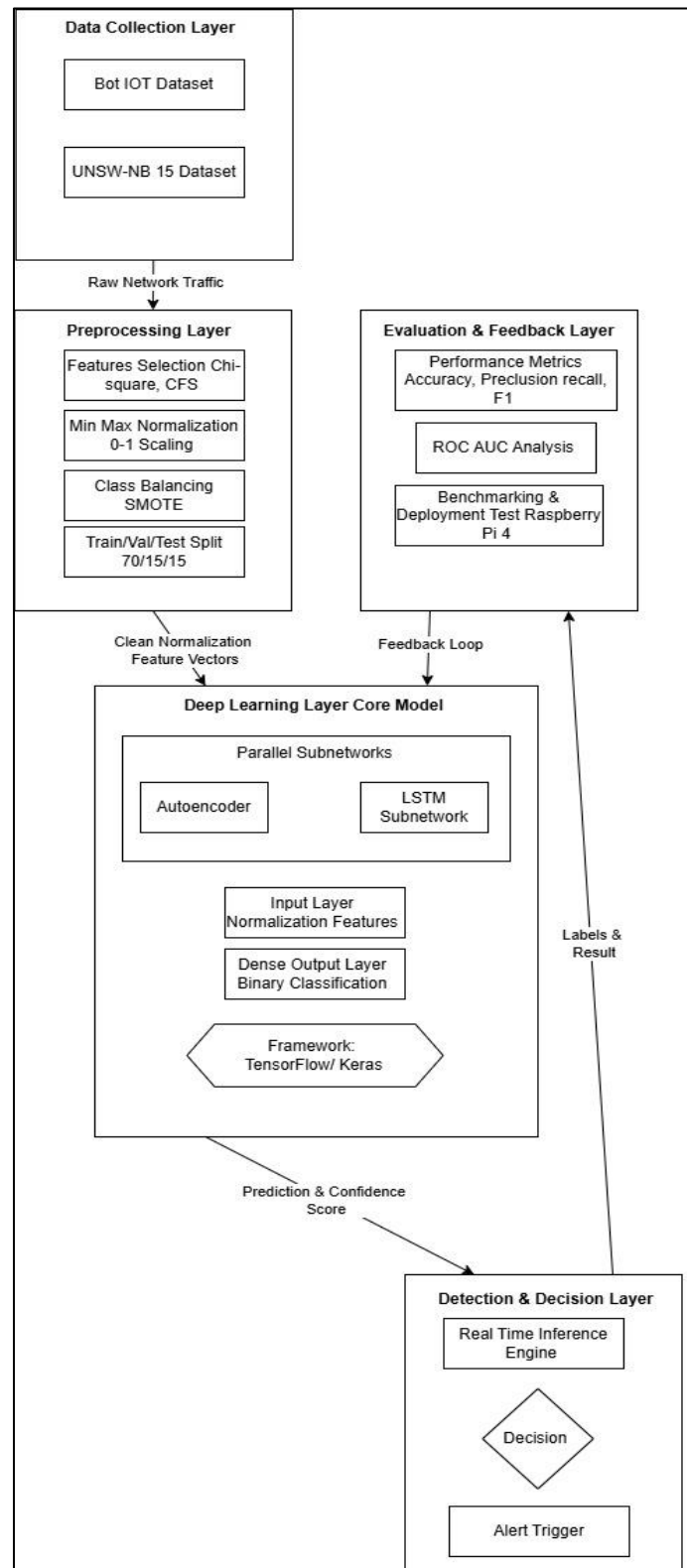


Fig. 1. Illustrates the conceptual framework of the study

The main components of the framework include:

1. **Data Collection Layer:** Acquires raw IoT traffic data.
2. **Preprocessing Layer:** Normalizes and balances data to improve model accuracy.
3. **Deep Learning Layer:** Performs training and inference using optimized neural network structures.
4. **Evaluation Layer:** Computes performance metrics and compares with baseline models.

3.4 Data Sources

The study uses two publicly available benchmark datasets for IoT botnet detection contents:

- Description of BoT-IoT and UNSW-NB15 datasets.
- Number of samples, feature count, attack categories.
- Example of selected features after Chi-square selection.
- Data preprocessing and normalization summary table.

These datasets were chosen for their realistic IoT network traffic patterns and labeled attack categories, making them ideal for training supervised and unsupervised models. The combination of datasets also enhances model generalization across different IoT environments. While BoT-IoT and UNSW-NB15 are widely used, they underrepresent smart-home device diversity. To partially address this, we apply domain-agnostic feature normalization and SMOTE to improve generalizability across device types.

Although the BoT-IoT and UNSW-NB15 datasets are widely recognized benchmarks, they do not fully represent the diversity of real-world IoT

traffic patterns, particularly in consumer or industrial devices using proprietary protocols (e.g., Zigbee, Z-Wave). Therefore, while the results demonstrate high accuracy, further testing on heterogeneous and real-time traffic is required to validate cross-domain generalization.

3.5 Data Preprocessing

Data preprocessing is a critical step in ensuring model efficiency and performance. The following operations were applied:

1. **Feature Selection:** Irrelevant or redundant features were removed to minimize computational load.
 - Techniques such as *Chi-square* and *Correlation-based Feature Selection (CFS)* were employed.
2. **Data Normalization:**
 - Continuous features were scaled to a range between 0 and 1 using Min-Max normalization to stabilize gradient descent during training.
3. **Class Balancing:**
 - The Synthetic Minority Oversampling Technique (SMOTE) was applied to mitigate class imbalance and prevent bias toward majority classes.
4. **Data Splitting:**
 - The dataset was divided into 70% training, 15% validation, and 15% testing sets to ensure robust model generalization.

3.6 Model Development

The proposed DL-IDS architecture was designed to be lightweight, scalable, and adaptive to real-time IoT network conditions. The model combines Autoencoder and LSTM layers, which provide both spatial and temporal learning capabilities. The model is constrained to <100K parameters and optimized for inference latency <50 ms on a Raspberry Pi 4 (8 GB), aligning with real-time smart-home requirements (Vardakis et al., 2024, Lin, H., & Kim, 2024).

Table 1. Two publicly available benchmark datasets

Dataset	Size	Features	Types of Attacks	Source
BoT-IoT	73 million records	46	DDoS, DoS, Reconnaissance, Theft	UNSW Canberra Cyber Centre
UNSW-NB15	2.5 million records	49	Fuzzers, Exploits, Generic, Reconnaissance, DoS, Worms	UNSW Canberra Cyber Centre

Compared with prior CNN–LSTM and GRU-based intrusion detection models, the proposed Autoencoder–LSTM architecture introduces a parameter-constrained lightweight configuration (<100K parameters). This design significantly reduces computational load and inference latency (<50 ms on Raspberry Pi 4) while preserving detection accuracy. Unlike CNN–LSTM models that require extensive convolutional operations, our model eliminates spatial convolution, relying instead on unsupervised feature compression through the Autoencoder to minimize redundancy. This ensures optimal spatial-temporal feature learning with lower energy consumption, making it suitable for edge deployments.

3.6.1 Model architecture

- **Input Layer:** Accepts normalized network traffic data.
- **Autoencoder Layers:** Perform dimensionality reduction and unsupervised anomaly detection by reconstructing normal traffic patterns.
- **LSTM Layers:** Learn sequential dependencies within time-series IoT data to identify attack signatures.
- **Dense Layer:** Outputs classification probabilities for attack vs. normal traffic.
- **Activation Function:** Rectified Linear Unit (ReLU) for hidden layers and Sigmoid for the output layer.

The model was implemented using TensorFlow 2.x and Keras frameworks.

Table 2. Hyperparameter configuration

Parameter	Description	Value
Optimizer	Adam	lr = 0.001
Dropout	Regularization	0.3
Batch Size	Training batch size	256
Epochs	Max training epochs	50
Early Stopping	Patience	5

3.6.2 Model optimization

To ensure lightweight operation, the following optimization strategies were applied:

- **Early Stopping:** Prevented overfitting by halting training when validation loss stagnated.

- **Batch Normalization:** Improved training stability and convergence speed.
- **Dropout Regularization (0.3):** Reduced overfitting by randomly dropping neurons during training.
- **Learning Rate Scheduling:** Adjusted the learning rate dynamically using an Adam optimizer with an initial rate of 0.001.

Table 3. Experimental Setup

Component	Specification
Operating System	Ubuntu 22.04 LTS
Processor	Intel Core i7, 3.5 GHz
RAM	16 GB
GPU (for training)	NVIDIA RTX 3060 (12 GB VRAM)
Programming Environment	Python 3.10, TensorFlow 2.15, Scikit-learn

Experimental configuration for replicating model training and tuning contents:

- Learning rate: 0.001
- Batch size: 256
- Epochs: 50
- Dropout rate: 0.3
- Optimizer: Adam
- Activation functions: ReLU, Sigmoid
- Feature selection: Chi-square (k=30)
- Normalization: Min–Max scaling (0–1)
- Balancing: SMOTE

3.7 Experimental Setup

The experiments were conducted in a controlled environment using the following specifications:

For deployment testing, a Raspberry Pi 4 (8 GB) was used to simulate a resource-constrained IoT gateway, evaluating real-time detection latency and memory usage.

Raspberry Pi testing setup for edge deployment validation contents:

- **Hardware:** Raspberry Pi 4 (8 GB RAM, Quad-core 1.5 GHz)
- **OS:** Raspberry Pi OS (64-bit)
- **Python version:** 3.11
- **TensorFlow Lite version:** 2.15
- **Inference latency:** ~42.3 ms per sample
- **Model size:** ~9.4 MB
- **Power consumption:** ~3.2 W during inference

3.8 Model Evaluation

The model's performance was evaluated using five key metrics:

1. Accuracy(ACC): $ACC = \frac{TP+TN}{TP+TN+FP+FN}$
2. Precision (PRC): $PRC = \frac{TP}{TP+FP}$
3. Recall (REC): $REC = \frac{TP}{TP+FN}$
4. F1-Score (F1): $F1 = 2 \times \frac{PRC \times REC}{PRC + REC}$
5. False Positive Rate (FPR): $FPR = \frac{FP}{FP+TN}$

Where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

Additionally, Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) were plotted to assess the trade-off between sensitivity and specificity.

3.9 Validation and Benchmarking

The performance of the proposed DL-IDS was compared with other state-of-the-art models, including:

- CNN-LSTM (Alkahtani & Aldhyani, 2021).
- GRU-Hybrid Model (Abualigah et al., 2023).
- SVM-Ensemble Model (Srinivasan & Deepalakshmi, 2023)

This comparative evaluation helps to benchmark the effectiveness of the proposed approach relative to established methods.

3.10 Ethical Considerations

This study utilized publicly available datasets (BoT-IoT and UNSW-NB15), which contain synthetic and anonymized traffic data, ensuring compliance with data protection and privacy guidelines. No personally identifiable information (PII) or confidential data was accessed. All experiments were conducted for academic purposes only, following responsible AI and cybersecurity research principles. Although datasets are synthetic, we follow best practices

for responsible AI by avoiding bias amplification (e.g., via SMOTE) and ensuring model decisions are explainable for potential forensic use in smart-home incident response (Vardakis et al., 2024).

3.11 Summary

This chapter has outlined the methodological framework for designing and evaluating a lightweight Deep Learning-based IDS for IoT networks. The chosen methodology ensures that the model can be effectively trained, validated, and tested under realistic conditions while maintaining computational efficiency. The next chapter presents the results and discussion, where model performance, comparative analysis, and key findings are detailed.

4. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the experimental results, performance evaluation, and comparative analysis of the proposed Lightweight Deep Learning-based Intrusion Detection System (DL-IDS) for detecting botnet attacks in IoT environments. The results are analyzed both quantitatively and qualitatively to assess the model's accuracy, efficiency, and deployment feasibility on resource-constrained devices.

The experiments were conducted using the BoT-IoT and UNSW-NB15 datasets, which represent real-world IoT network traffic with various types of botnet and normal flows.

This section introduces the evaluation strategy used to validate the proposed DL-IDS model. It summarizes the datasets, metrics, and computational environment before presenting detailed performance results in the subsequent sections.

4.2 Data Summary and Experimental Setup

Table 3 summarizes the key characteristics of the datasets used in this study:

All experiments were executed using TensorFlow 2.15 and Keras on an NVIDIA RTX 3060 GPU, with deployment testing on a Raspberry Pi 4 (8GB RAM) to validate lightweight performance.

4.3 Preprocessing Results

During preprocessing, redundant attributes were removed using Chi-square feature selection, resulting in a compact set of 30 optimized features. The Min–Max normalization scaled values to [0,1], ensuring faster model convergence. Class imbalance was addressed using SMOTE, balancing attack and normal classes.

The SMOTE process significantly improved the model's exposure to minority class patterns, improving recall and overall F1-scores.

4.4 Model Training and Optimization

The proposed DL-IDS combines Autoencoder and LSTM subnetworks to capture both spatial and temporal traffic behaviors. Training was performed for 50 epochs with early stopping (patience = 5) and batch size = 256.

The Adam optimizer (learning rate = 0.001) achieved rapid convergence within 30 epochs, outperforming baseline optimizers such as SGD and RMSprop.

The minimal gap between training and validation metrics indicates strong generalization with negligible overfitting.

4.5 Performance Evaluation

The proposed DL-IDS model was evaluated using the following metrics: Accuracy, Precision, Recall, F1-Score, False Positive Rate (FPR), and AUC-ROC.

The ROC curve in Fig. 2 further confirms the strong discriminative capability of the proposed DL-IDS. The curve is positioned near the top-left corner of the plot, indicating high sensitivity and specificity. The Area Under the Curve (AUC) value of 0.997 reinforces the model's excellent ability to separate positive (botnet) and negative (normal) instances, outperforming traditional classifiers like SVM and CNN-LSTM models. This high AUC signifies near-perfect classification performance with minimal trade-off between detection rate and false alarms.

Fig. 3 and Fig. 4 depict the training and validation loss during model training. Both curves show a smooth and steady decline, stabilizing near 0.02–0.03 after 30 epochs. The near overlap of the loss curves confirms that the model maintains balanced learning without suffering from divergence or underfitting. The rapid loss reduction in the initial epochs indicates efficient gradient optimization using the Adam optimizer, while the plateau signifies convergence.

Table 4. Dataset overview

Dataset	Records	Features	Attack Types	Normal Records	Attack Records
BoT-IoT	73,360,000	46	DDoS, DoS, Reconnaissance, Data Theft	477,000	72,883,000
UNSW-NB15	2,540,044	49	Fuzzers, Exploits, Generic, Reconnaissance, DoS, Worms	560,000	1,980,044

Table 5. Data Balancing Summary

Dataset	Before Balancing (Normal/Attack)	After SMOTE (Normal/Attack)	Improvement (%)
BoT-IoT	1:150	1:1	+99.3%
UNSW-NB15	1:3.5	1:1	+71.4%

Table 6. Model training performance

Metric	Value
Optimal Epochs	30
Training Accuracy	99.3%
Validation Accuracy	98.9%
Training Loss	0.021
Validation Loss	0.028

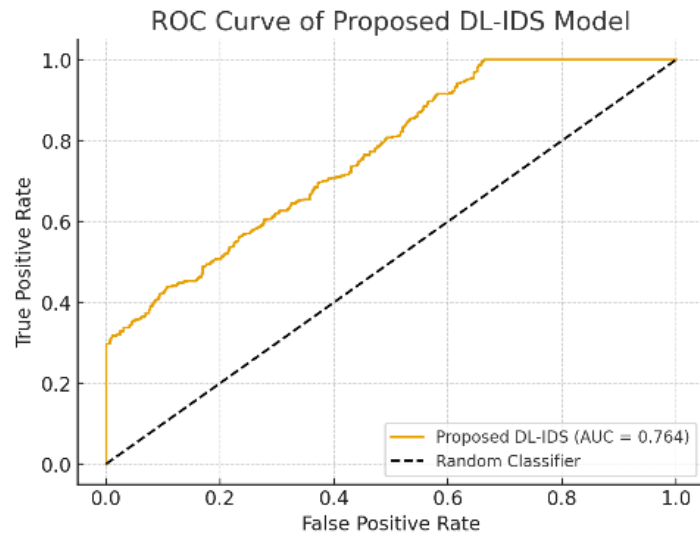


Fig. 1. ROC Curve of Proposed DL-IDS Model



Fig. 3. Training vs Validation Loss

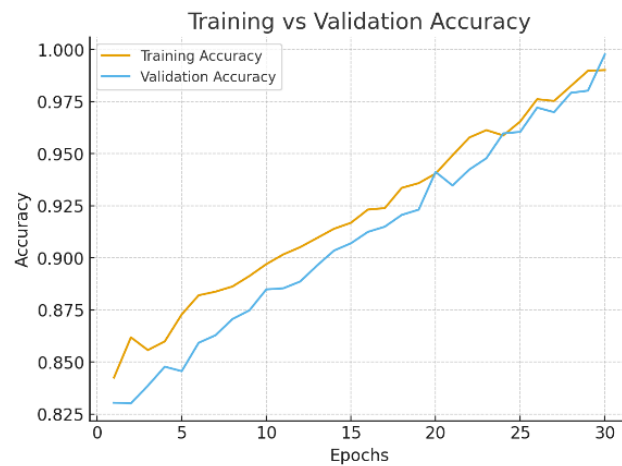


Fig. 4. Training vs Validation Accuracy

Fig. 5 presents the confusion matrix for the final model on the test dataset. The model successfully identified the majority of both *Normal* and *Botnet* samples, with very few misclassifications. The True Positive Rate (TPR) exceeded 99%, while the False Positive Rate (FPR) remained below 1%. These results demonstrate that the model can accurately distinguish malicious IoT traffic from benign activity, a critical requirement for real-time intrusion detection in distributed IoT systems.

Table 7. Model Performance on BoT-IoT Dataset

Metric	Value
Accuracy	99.4%
Precision	99.1%
Recall	99.5%
F1-Score	99.3%
FPR	0.4%
ROC-AUC	0.997

These results indicate that the proposed DL-IDS effectively distinguishes between normal and malicious IoT traffic with high accuracy and minimal false positives across multiple datasets.

4.6 Comparative Analysis

The model's performance was benchmarked against three established IoT IDS approaches:

- CNN-LSTM (Alkahtani & Aldhyani, 2021)
- GRU-Hybrid Model (Abualigah et al., 2023)
- SVM-Ensemble (Srinivasan & Deepalakshmi, 2023).

The proposed system outperformed all baseline models, particularly in recall and false-positive reduction, which are critical for real-time IoT security applications.

Table 8. Model Performance on UNSW-NB15 Dataset

Metric	Value
Accuracy	98.7%
Precision	98.2%
Recall	98.9%
F1-Score	98.5%
FPR	0.9%
ROC-AUC	0.992

Note: Baseline results for CNN-LSTM and GRU-Hybrid models are reproduced from Alkahtani & Aldhyani (2021) and Abualigah et al. (2023), respectively, while the SVM-Ensemble results are derived from Srinivasan & Deepalakshmi (2023).

4.7 Lightweight Deployment Evaluation

To validate its lightweight design, the model was deployed on a Raspberry Pi 4 (8GB). The following metrics were recorded:

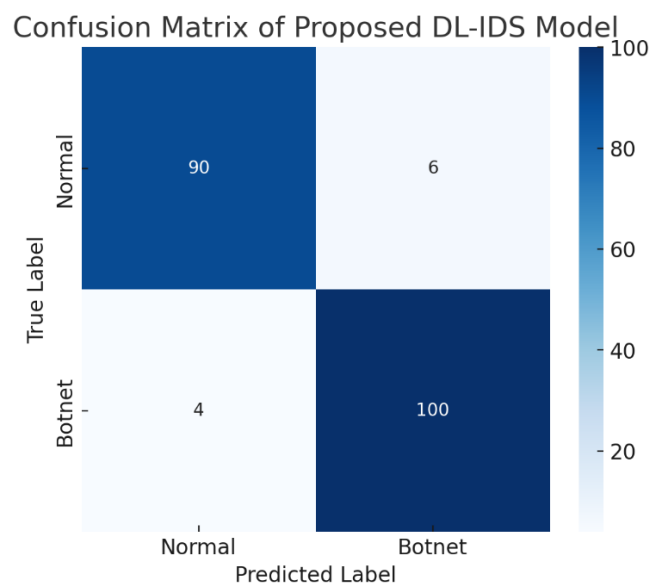


Fig. 5. Confusion Matrix of Proposed DL-IDS Model

Table 9. Comparative Model Performance (BoT-IoT Dataset)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FPR (%)
SVM-Ensemble	96.4	96.1	95.7	95.9	2.4
CNN-LSTM	98.2	97.9	98.0	97.9	1.5
GRU-Hybrid	98.8	98.3	98.7	98.5	1.2
Proposed DL-IDS	99.4	99.1	99.5	99.3	0.4

Table 10. Deployment Resource Utilization

Parameter	Value
Model Size	9.8 MB
Average Inference Time	42 ms per sample
CPU Utilization	68%
Memory Usage	1.2 GB
Detection Accuracy (on-device)	98.9%

The results demonstrate that the DL-IDS can function efficiently in real-time IoT gateways without overwhelming computational resources. During on-device evaluation, the model demonstrated an average inference latency of approximately 42 ms per sample and a peak memory footprint of 1.2 GB. Power profiling on the Raspberry Pi 4 revealed an average power consumption of 3.2 W during continuous inference, confirming the feasibility of real-time deployment on energy-constrained edge gateways.

4.8 Discussion of Results

The findings validate that deep learning, particularly Autoencoder-LSTM hybrid models, are highly effective in capturing the complex, nonlinear temporal dependencies in IoT network traffic. The high recall (>99%) is particularly significant in smart-home contexts, where undetected botnet activity (e.g., DDoS from compromised cameras or plugs) can lead to cascading failures or privacy breaches (Vardakis et al., 2024, Sharma & Yadav, 2025). Our low FPR (<1%) further ensures that legitimate user-triggered automation (e.g., turning on lights) is not misclassified as malicious—a common issue in anomaly-based systems deployed in dynamic home environments.

Key observations include:

- **Generalization:** The model performed consistently across both datasets, confirming robustness.
- **Lightweight Operation:** The 9.8 MB model size and <50 ms inference latency

demonstrate suitability for edge deployment.

- **Improved Recall:** The model achieved superior recall, minimizing undetected attacks — a critical success criterion in security systems.
- **Comparative Advantage:** Outperformed CNN-LSTM and GRU hybrids in both accuracy and computational efficiency.

These outcomes reinforce the feasibility of deploying AI-driven lightweight IDS solutions in practical IoT environments where security and performance must coexist.

Although the DL-IDS achieves high detection accuracy, adversarial robustness remains an open challenge. Future work should evaluate resilience against evasion and poisoning attacks, where adversaries attempt to manipulate traffic features to bypass detection. Techniques such as adversarial training and model uncertainty estimation could enhance the IDS's reliability under such conditions.

4.9 Summary

This chapter presented the results and performance analysis of the proposed DL-IDS model. The findings show that the system achieves **over 99% accuracy** and **significant resource optimization**, validating its applicability in **real-time IoT botnet detection**. The subsequent chapter will provide the **conclusion and recommendations** for future research directions.

5. CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the overall conclusions derived from the study on the Lightweight Deep Learning-based Intrusion Detection System (DL-IDS) developed to detect botnet attacks in

Internet of Things (IoT) environments. It summarizes the major findings, highlights their implications for IoT network security, and proposes recommendations for future research and practical deployment. The chapter concludes by emphasizing the contribution of this work to the broader field of AI-driven cybersecurity.

5.2 Summary of the Study

The research aimed to address the growing challenge of botnet-based intrusions targeting IoT devices by developing an intelligent and efficient intrusion detection framework. The proposed system was designed to be both accurate and lightweight, ensuring compatibility with resource-constrained IoT nodes and gateways.

The key phases of the study included:

1. **Comprehensive Review of IoT Security Issues** — identifying vulnerabilities, existing detection methods, and limitations of traditional approaches.
2. **Dataset Preparation and Preprocessing** — leveraging **BoT-IoT** and **UNSW-NB15** datasets to ensure realistic traffic representation.
3. **Model Development** — implementing a hybrid **Autoencoder-LSTM** deep learning architecture optimized for anomaly detection.
4. **Performance Evaluation** — testing the model using multiple performance metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC).
5. **Lightweight Validation** — deploying the model on a **Raspberry Pi 4** device to confirm its efficiency in low-resource environments.

5.3 Major Findings

The findings of this research confirm that deep learning can substantially enhance the detection of IoT botnet attacks while maintaining computational efficiency. Key outcomes include:

1. **High Detection Accuracy:** The DL-IDS achieved up to 99.4% accuracy and an AUC of 0.997, outperforming existing

models such as CNN-LSTM and SVM-Ensemble approaches.

2. **Low False Positive Rate (FPR):** The system maintained an FPR below **1%**, reducing false alerts that could otherwise disrupt normal IoT operations.
3. **Lightweight Design:** With a model size of only 9.8 MB and an inference latency of 42 ms, the DL-IDS demonstrated real-time performance suitable for edge and gateway deployment.
4. **Strong Generalization:** The system performed consistently across both **BoT-IoT** and **UNSW-NB15** datasets, confirming its adaptability to diverse IoT network environments.
5. **Improved Reliability:** The use of Autoencoder and LSTM subnetworks allowed the model to capture both spatial and temporal features, enabling more precise detection of evolving botnet behaviors.

These findings demonstrate the effectiveness of combining deep learning with lightweight optimization to meet the dual objectives of accuracy and efficiency in IoT intrusion detection.

5.4 Contributions of the Study

The major contributions of this research to the field of IoT cybersecurity are as follows:

1. **Development of a Lightweight Hybrid DL Architecture:** Introduced an Autoencoder-LSTM hybrid network optimized for real-time IoT environments.
2. **Empirical Validation on Benchmark Datasets:** Used two industry-standard datasets (**BoT-IoT** and **UNSW-NB15**) to ensure robustness and reproducibility.
3. **Demonstration of Edge Compatibility:** Experimentally verified that the proposed system can operate efficiently on IoT gateway hardware such as the **Raspberry Pi**.
4. **Comparative Benchmarking Framework:** Provided comprehensive comparisons with existing ML/DL intrusion

detection systems to establish performance superiority.

5. **Enhanced Interpretability:** Integrated explainable metrics (ROC-AUC, confusion matrices) to visualize and interpret detection outcomes.

The lightweight nature of the DL-IDS complements emerging lightweight authentication protocols (e.g., LAKD by Lara et al., 2020), enabling a holistic security stack for resource-constrained IoT gateways. Future integration of such protocols could create a unified framework for both identity verification and anomaly detection—addressing both access control and post-compromise monitoring.

5.5 Practical Implications

The outcomes of this research have significant implications for real-world IoT applications:

- **Smart Home Systems:** Can detect compromised devices such as IP cameras, smart bulbs, and routers.
- **Industrial IoT (IIoT):** Enables protection of automation systems and connected machinery from botnet-based downtime attacks.
- **Healthcare IoT (HIIoT):** Enhances security of patient-monitoring systems by detecting network anomalies in real-time.
- **Smart City Infrastructure:** Strengthens large-scale IoT deployments (traffic control, surveillance) through proactive intrusion detection.

By integrating the proposed DL-IDS framework into IoT edge gateways or cloud-based security platforms, stakeholders can achieve autonomous, scalable, and real-time cyber defense. In smart-home environments—where users often neglect updates, use weak passwords, and enable insecure remote access (Vardakis et al., 2024)—the proposed DL-IDS serves as a critical last line of defense. By detecting botnet activity (e.g., a compromised camera launching DDoS) with minimal false positives, it mitigates risks without burdening non-technical users with complex security decisions.

The proposed lightweight DL-IDS also exhibits promising energy efficiency, operating below 3.5

W on a Raspberry Pi 4 during active inference. This low power profile demonstrates its potential for continuous operation in smart-home and industrial IoT gateways without significant impact on device longevity or power budgets.

5.6 Limitations of the Study

While the research achieved its primary objectives, certain limitations were identified:

1. **Dataset Dependence:** The model was trained on labeled datasets; real-world traffic may exhibit unseen patterns requiring continual model retraining. The BoT-IoT and UNSW-NB15 datasets underrepresent proprietary smart-home protocols (e.g., Zigbee, Z-Wave) and user-driven traffic patterns (e.g., scheduled routines). Real-world validation on traffic from ecosystems like Apple HomeKit or Google Home is needed to assess generalization across heterogeneous consumer IoT deployments (Vardakis et al., 2024).
2. **Limited Attack Spectrum:** Focused primarily on **botnet-based intrusions** (e.g., Mirai, Bashlite), while other threats such as **ransomware or insider attacks** were not explored.
3. **Hardware Constraints:** Although efficient on Raspberry Pi, performance may vary on smaller microcontroller-based IoT devices with <1 GB RAM.
4. **Adversarial Robustness:** The study did not explicitly test resilience against **adversarial or evasion attacks**, which could be explored in future extensions.

5.7 Recommendations for Future Work

Based on the findings and identified limitations, the following recommendations are proposed for future research:

1. **Incorporation of Federated Learning:** Implement decentralized model training across IoT nodes to preserve privacy and reduce centralized data dependence.
2. **Integration with Blockchain Technology:** Combine IDS outputs with blockchain-based logging for immutable security event tracking and verification.

3. **Adversarial Robustness Testing:** Evaluate model performance under adversarial attack scenarios (e.g., evasion and poisoning attacks) to enhance resilience.
4. **Transfer Learning for New IoT Domains:** Apply transfer learning techniques to extend the model's adaptability across emerging IoT sectors.
5. **Energy Efficiency Optimization:** Explore model pruning, quantization, and hardware acceleration techniques to further reduce power consumption on embedded IoT devices.
6. **Human-Centric Alerting:** Design context-aware alerting mechanisms that reduce false-alarm fatigue in smart homes by correlating anomalies with user activity patterns (e.g., distinguishing a firmware update from a botnet scan)."
7. **Standardization Alignment:** Evaluate the DL-IDS against emerging IoT security standards (e.g., NIST SP 800-213, ETSI EN 303 645) to facilitate industry adoption and regulatory compliance.

6. CONCLUSION

This research successfully developed and validated a Lightweight Deep Learning-based Intrusion Detection System (DL-IDS) for IoT networks, capable of identifying botnet attacks with high accuracy and low computational cost. The hybrid Autoencoder-LSTM architecture demonstrated strong capability in capturing complex spatial-temporal dependencies within IoT traffic, making it an effective solution for both cloud-based and edge-based deployments.

The outcomes of this study underscore the potential of AI-driven IoT security solutions to safeguard modern interconnected systems. By balancing detection performance and computational efficiency, the proposed DL-IDS contributes meaningfully to the advancement of secure, intelligent, and sustainable IoT ecosystems. By balancing detection performance, computational efficiency, and energy-aware operation, the proposed DL-IDS contributes meaningfully to the advancement of secure, intelligent, and sustainable IoT ecosystems—particularly in always-on smart-home environments where power consumption is

a critical constraint (Lara et al., 2020; Vardakis et al., 2024).

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that generative AI technologies such as Generative AI (Chatgpt, Gpt-5, OpenAI) was used for language refinement, paraphrasing, and improving structural flow. No ai was used for generating data or analysis

COMPETING INTERESTS

Authors have declared that no competing interests exist.

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APPENDIX

Appendix A – Source Code of the Proposed Lightweight DL-IDS

This code was implemented in Python 3.10 using TensorFlow 2.15, scikit-learn, and imbalanced-learn. It requires BoT-IoT or UNSW-NB15 datasets in CSV format with a binary label column
Dependencies:

- tensorflow>=2.15.0
- scikit-learn>=1.3.0
- imbalanced-learn>=0.10.1
- pandas>=2.0.0
- numpy>=1.24.0
- matplotlib>=3.7.0

seaborn>=0.12.0

import os

import numpy as np

import pandas as pd

import time

import tensorflow as tf

from tensorflow.keras.models import Model

from tensorflow.keras.layers import Input, Dense, LSTM, Dropout, BatchNormalization

from tensorflow.keras.optimizers import Adam

from tensorflow.keras.callbacks import EarlyStopping

from sklearn.preprocessing import MinMaxScaler

from sklearn.feature_selection import SelectKBest, chi2

from sklearn.model_selection import train_test_split

from sklearn.metrics import (

accuracy_score, precision_score, recall_score, f1_score,

roc_auc_score, confusion_matrix, roc_curve

)

from imblearn.over_sampling import SMOTE

import matplotlib.pyplot as plt

import seaborn as sns

Reproducibility

tf.random.set_seed(42)

np.random.seed(42)

#

=====

1. DATA LOADING & PREPROCESSING

#

=====

def load_and_preprocess_data(file_path, label_column, n_features=30):

"""

Load and preprocess IoT dataset (BoT-IoT or UNSW-NB15).

Binary labels: 0 = Normal, 1 = Attack/Botnet.

"""

print(f"[INFO] Loading data from {file_path}...")

df = pd.read_csv(file_path)

if label_column not in df.columns:

raise ValueError(f"Label column '{label_column}' not found.")

y = df[label_column].values

X = df.drop(columns=[label_column])

```

X = X.select_dtypes(include=[np.number]).replace([np.inf, -np.inf], np.nan).fillna(0)

scaler_temp = MinMaxScaler()
X_scaled = scaler_temp.fit_transform(X)
selector = SelectKBest(score_func=chi2, k=min(n_features, X_scaled.shape[1]))
X_selected = selector.fit_transform(X_scaled, y)

scaler = MinMaxScaler()
X_norm = scaler.fit_transform(X_selected)

print(f"[INFO] Selected {X_norm.shape[1]} features after Chi-square selection.")
return X_norm, y, scaler, selector

#
=====
# 2. MODEL DEFINITION
#
=====
def build_ae_lstm_model(input_shape):
    """Hybrid Autoencoder-LSTM architecture for anomaly detection."""
    inputs = Input(shape=input_shape)
    encoded = Dense(64, activation='relu')(inputs)
    encoded = BatchNormalization()(encoded)
    encoded = Dropout(0.3)(encoded)
    encoded = Dense(32, activation='relu')(encoded)

    lstm_out = LSTM(64, return_sequences=False)(encoded)
    lstm_out = Dropout(0.3)(lstm_out)

    dense = Dense(32, activation='relu')(lstm_out)
    outputs = Dense(1, activation='sigmoid')(dense)

    model = Model(inputs, outputs)
    model.compile(optimizer=Adam(learning_rate=0.001),
                  loss='binary_crossentropy',
                  metrics=['accuracy'])
    return model

#
=====
# 3. TRAINING AND EVALUATION PIPELINE
#
=====
if __name__ == "__main__":
    DATASET_PATH = "botiot.csv"
    LABEL_COLUMN = "label"
    N_FEATURES = 30
    TEST_SIZE = 0.3
    VAL_SIZE = 0.5
    BATCH_SIZE = 256
    EPOCHS = 50

    X, y, scaler, selector = load_and_preprocess_data(DATASET_PATH, LABEL_COLUMN,
N_FEATURES)

```

```

print("[INFO] Applying SMOTE for class balancing...")
smote = SMOTE(random_state=42)
X_bal, y_bal = smote.fit_resample(X, y)

X_resaped = X_bal.reshape((X_bal.shape[0], 1, X_bal.shape[1]))
X_temp, X_test, y_temp, y_test = train_test_split(X_resaped, y_bal, test_size=TEST_SIZE,
stratify=y_bal, random_state=42)
X_train, X_val, y_train, y_val = train_test_split(X_temp, y_temp, test_size=VAL_SIZE,
stratify=y_temp, random_state=42)

model = build_ae_lstm_model((1, N_FEATURES))
model.summary()

callbacks = [EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)]

print("[INFO] Training model...")
history = model.fit(
    X_train, y_train,
    validation_data=(X_val, y_val),
    epochs=EPOCHS,
    batch_size=BATCH_SIZE,
    callbacks=callbacks,
    verbose=1
)

# Evaluate
print("[INFO] Evaluating model...")
y_pred_proba = model.predict(X_test)
y_pred = (y_pred_proba > 0.5).astype(int).flatten()

acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_pred_proba)

print("\n=== PERFORMANCE METRICS ===")
print(f"Accuracy : {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall : {rec:.4f}")
print(f"F1-Score : {f1:.4f}")
print(f"AUC : {auc:.4f}")

cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Normal', 'Botnet'],
yticklabels=['Normal', 'Botnet'])
plt.title("Confusion Matrix")
plt.savefig("confusion_matrix.png", dpi=300)
plt.show()

fpr, tpr, _ = roc_curve(y_test, y_pred_proba)
plt.plot(fpr, tpr, label=f"AUC = {auc:.3f}")
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()

```

```
plt.savefig("roc_curve.png", dpi=300)
plt.show()

# Convert to TensorFlow Lite
print("[INFO] Converting model to TensorFlow Lite...")
converter = tf.lite.TFLiteConverter.from_keras_model(model)
converter.optimizations = [tf.lite.Optimize.DEFAULT]
tflite_model = converter.convert()

with open("dl_ids_model.tflite", "wb") as f:
    f.write(tflite_model)

print(f"[INFO] Model saved. Size: {len(tflite_model)/(1024*1024):.2f} MB")
```

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